

Signal-equivalence: testable dose-reduction claims for medical imaging

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Abstract

Claims about reduced-signal medical imaging — “50% dose reduction in CT,” “4× acceleration in MRI,” “25%-activity PET” — are routinely reported as single scalar percentages without specifying the diagnostic task, the patient subpopulation, the equivalence margin, or the statistical confidence over which the claim is meant to hold. The resulting literature is neither comparable across studies nor actionable for clinical decision-making, and the gap is widening as reconstruction methods proliferate.

We introduce **signal-equivalence**, a probabilistic framework that converts every such claim into a 5-tuple credential $(r, T, \varepsilon, \alpha, \Pi)$: a reconstruction method is signal-equivalent to a named reference method at signal-reduction ratio r , task T , equivalence margin ε , and significance level α over patient subpopulation Π if and only if its expected task performance on reduced-signal scans is within ε of the reference method’s performance on full-signal scans, with confidence at least $1 - \alpha$. The framework is modality-general by construction — a single definition covers CT tube-current reduction, MRI k -space undersampling, and PET activity reduction — and inherits its statistical machinery from the test-of-equivalence procedures that the clinical-trial community has used for forty years but that medical-imaging methods papers have not adopted as a reporting standard.

We provide a paired-bootstrap estimator with finite-sample coverage guarantees, a closed-form sample-size formula, and an open-source, pip-installable Python library (`pwm_dose_equivalence`) that produces credentials from any reconstruction method’s image outputs. We validate the framework with three worked examples spanning CT, MRI, and PET, in each case using the same single API call. To prevent silent benchmark drift, each credential is content-addressed by the SHA-256 hash of the framework definition it was issued against, so a reader given only a credential JSON can recover the exact framework version and recompute the verdict bit-for-bit.

Adopted as a reporting standard, signal-equivalence converts marketing claims into reviewable scientific propositions and enables both human reviewers and automated agents to verify those claims against published evidence.

1 Introduction

A central question in modern medical imaging is whether a new reconstruction method delivers diagnostically equivalent images at reduced patient exposure or scan time. The question takes a specific form in each modality: in computed tomography it is asked of reduced tube current (“low-dose CT”); in magnetic resonance imaging of k -space undersampling (“accelerated MRI”); in positron emission tomography of reduced injected radiotracer activity (“low-activity PET”).

Across modalities, the dominant evaluation protocol is to report a single scalar metric — peak signal-to-noise ratio (PSNR), structural similarity (SSIM), root-mean-squared error (RMSE) — alongside an unqualified claim such as “50% dose reduction” or “4× acceleration.”

This reporting practice has three interconnected failures.

Failure 1: undefined task. A reconstruction that preserves PSNR may smooth out the diagnostic feature it was supposed to preserve (Kim et al., 2019; Christianson et al., 2015). Without specifying the downstream clinical task — lung-nodule detection, lesion segmentation, organ quantification — the equivalence claim is unanchored. A method may be “dose-equivalent” for lung-nodule detection while being “dose-incompatible” for ground-glass opacity characterization on the same scan.

Failure 2: undefined subpopulation. Methods optimized on adult chest scans frequently degrade on pediatric or oncology-followup populations because the noise statistics, anatomical priors, and lesion-prevalence distributions differ (U.S. Food and Drug Administration, 2024; Larson et al., 2018). An equivalence claim that does not specify the patient distribution over which it holds cannot be transferred to a new clinical context without re-validation.

Failure 3: undefined statistics. “Equivalent” as a binary verdict, without a margin ε and a confidence level α , is not testable. Test-of-equivalence statistics — two one-sided tests (TOST), non-inferiority trial design — have been the standard in pharmacokinetics and clinical-trial design for forty years (Schuirmann, 1987; Piaggio et al., 2012), but have not been adopted as a methodological standard in medical-imaging methods papers.

We argue these three failures share a single solution: every equivalence claim should be a structured, probabilistic, task-conditional, subpopulation-conditional 5-tuple credential. We call this the *signal-equivalence framework*. The framework is intentionally modality-general: rather than producing one definition for CT (with parameter “mA ratio”), a second for MRI (“acceleration factor”), and a third for PET (“activity ratio”), we adopt a single signal-reduction parameter $r \in (0, 1]$ and a modality-specific signal-reduction operator T_r . The framework is then identical across modalities, while each modality’s instantiation of T_r encodes its physics.

Relation to existing equivalence statistics. The statistical machinery is not new: test-of-equivalence procedures such as Schuirmann’s two one-sided tests (TOST) (Schuirmann, 1987) and non-inferiority trial design (Piaggio et al., 2012) have served bioequivalence and clinical-trial endpoint reporting for nearly forty years, and CONSORT (Schulz et al., 2010) requires that every randomized-trial publication report a margin and a confidence level. Medical-imaging *methods* papers have not adopted this discipline. The signal-equivalence framework can be read as TOST applied to a paired-on-patient performance-difference estimator, with the four extra components — explicit task T , explicit subpopulation Π , explicit signal-reduction operator T_r , and a content-addressed credential schema — that are needed for the medical-imaging-methods setting but absent from the pharmacokinetic source material.

Relation to task-based image-quality assessment. A separate, medical-imaging-internal lineage of work has long argued that image quality should be measured against a clinical task, not against scalar fidelity metrics — the canonical statement is Barrett (1990), and the textbook reference is Barrett and Myers (2013), which formalizes the Hotelling and channelized-Hotelling observer families and the ideal-observer figure-of-merit. AAPM Task Group 233 (Samei et al., 2019) translates this lineage into a CT-specific performance-evaluation standard built around model

observers on standardized phantoms. Signal-equivalence is *complementary to*, not a replacement for, task-based image-quality assessment (TB-IQ). TB-IQ scores a single reconstruction-system configuration on a fixed phantom or test set, using a model observer as the figure of merit; it answers “how good is system S ?” Signal-equivalence tests a pairwise equivalence claim between two methods at an explicit signal-reduction ratio over a named patient subpopulation; it answers “does method M at reduced signal match reference method $\mathcal{M}_{\text{ref}}$ at full signal, by how much, and with what confidence?” The two are usefully composed: any TB-IQ figure of merit that produces a real-valued score per case — including the channelized-Hotelling observer of [Barrett and Myers \(2013\)](#) — is a valid choice for $P(T(\cdot), g(\cdot))$ inside a signal-equivalence credential. What signal-equivalence adds on top of the TB-IQ lineage is (a) explicit equivalence-margin and confidence-level reporting, (b) explicit subpopulation as a measure, (c) a signal-reduction operator T_r that ties the claim to a specific physical reduction, and (d) a content-addressed credential format. None of these is present in the TB-IQ standard, and none of the TB-IQ contributions is duplicated here.

We make four separable contributions, each independently testable and adoptable.

1. **A modality-general formalization of equivalence reporting for image-based clinical tasks.** The framework’s mathematical kernel is a paired-on-patient equivalence test; what is novel is the type-signature discipline that ties an equivalence claim to a specific signal-reduction operator T_r defined per modality (Section 4.1), a task with explicit ground-truth protocol, a subpopulation expressed as a measure, and either an aggregate or per-patient formulation. The same definition covers CT tube-current reduction, MRI k -space undersampling, and PET activity reduction — a unification not available in any existing reporting standard or statistics package.
2. **An estimator with finite-sample guarantees and pre-registered sample-size targets** (Section 4.3). The paired bootstrap is conservative for AUC near its boundary; we provide a bias-corrected accelerated (BCa) variant and an exact DeLong test variant for the AUC case, both wired into the same API, with automatic regime detection that warns when the percentile CI is anti-conservative.
3. **An open-source, pip-installable, type-annotated, CI-tested Python library** (`pwm_dose_equivalence`, Apache 2.0) with worked-example notebooks for three modalities (Section 4.4, Section 4.5). Reviewers can install the library, run the notebooks, and reproduce every result in this paper.
4. **A content-addressed credential schema** that ties each published credential to the SHA-256 hash of the exact framework definition it was issued against (Section 4.6). A reader holding only a credential JSON can recover the precise framework version, re-run the same library on the same test set, and verify the verdict bit-for-bit — eliminating a class of benchmark-drift artifacts that has undermined longitudinal comparison in medical imaging ([Wang et al., 2020](#)). Content-addressing is the methodological substance of the contribution; the choice of where to host the hash (local file, package registry, or public chain) is a deployment detail discussed in Section 4.6.

A reader can adopt contribution 1 (the framework) without 4 (the registry); contribution 2 (the estimator) without 3 (the library); contribution 3 (the library) without 1 (by writing a different formal definition that the library compiles to). The four contributions are deliberately layered so that each subsequent one strengthens but does not depend on the previous one.

2 Results

Framework overview. We define an *imaging modality* as a triple $\mathcal{M} = (\mathcal{X}, \mathcal{S}, \mathcal{Y})$ consisting of a patient space $\mathcal{X}$, an acquisition space $\mathcal{S}$, and a reconstruction space $\mathcal{Y}$, together with a reference acquisition operator $S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{S})$ (probabilistic; acquisition is noisy). A *signal-reduction operator at level $r \in (0, 1]$* is a map $T_r : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(\mathcal{S})$ with the interpretation that $T_r(S_{\text{ref}}(x))$ is the raw measurement distribution one would obtain at reference-level acquisition scaled by r (Section 4.1 formalizes this; the modality-specific instantiations are in Table 6).

The signal-equivalence credential for a reconstruction method M against a reference method $\mathcal{M}_{\text{ref}}$ at the level $(r, T, \varepsilon, \alpha, \Pi)$ asserts that M 's expected task performance on reduced-signal scans is within ε of $\mathcal{M}_{\text{ref}}$'s expected task performance on full-signal scans, with confidence $\geq 1 - \alpha$. The credential is computed via a paired bootstrap over patient draws from Π (Section 4.3); the verdict is one of PASS, FAIL, or INDETERMINATE (CI straddles the equivalence band).

Worked example 1: CT lung-nodule detection at 25% dose. On the public LIDC-IDRI (Armato III et al., 2011) and AAPM-2016 (AAPM, 2016) cohorts and the multi-vendor PWM-LDCT dataset (to be confirmed, 2027a), we computed signal-equivalence credentials for three baseline reconstruction methods (RED-CNN (Chen et al., 2017), a transformer-based reconstruction, a diffusion-based reconstruction) and the open-source PWM reference method (to be confirmed, 2027b) at $r = 0.25$. Performance metric: lung-nodule detection AUC at fixed FPR = 0.1, with ground truth from majority-vote radiologist annotation. Results are summarized in Table 1.

[TODO: TABLE: Signal-equivalence credentials for four CT reconstruction methods at $r = 0.25$, task = lung-nodule detection AUC]

Table 1: Signal-equivalence credentials for four CT reconstruction methods at $r = 0.25$, task = lung-nodule detection AUC (placeholder; populate when data lands).

Worked example 2: accelerated MRI knee-meniscus segmentation at $4\times$. On the fastMRI knee dataset (Zbontar et al., 2020), we applied the framework with variable-density Cartesian undersampling at $R = 4$ (equivalently $r = 0.25$); the mask family is recorded as part of Π so the 5-tuple shape is preserved (Section 4.1). Performance metric: meniscus segmentation Dice. Results in Table 2.

[TODO: TABLE: Signal-equivalence credentials for MRI reconstruction methods at $r = 0.25$ ($4\times$ acceleration), task = knee-meniscus segmentation Dice]

Table 2: Signal-equivalence credentials for MRI reconstruction methods at $r = 0.25$ ($4\times$ acceleration), task = knee-meniscus segmentation Dice (placeholder; populate when data lands).

Worked example 3: PET phantom quantification at 25% activity. On the NEMA NU-2 IQ phantom (NEMA, 2018) at full and reduced-activity acquisitions, we measured contrast-recovery coefficients for the six standard sphere sizes. Results in Table 3.

[TODO: TABLE: Signal-equivalence credentials for PET reconstruction methods at $r = 0.25$ (25% injected activity), task = contrast recovery per sphere size]

Table 3: Signal-equivalence credentials for PET reconstruction methods at $r = 0.25$ (25% injected activity), task = contrast recovery per sphere size (placeholder; populate when data lands).

Cross-modality consistency. The same Python API call signature (Section 4.4) produces a credential for each modality:

```

from pwm_dose_equivalence import signal_equivalence_credential
cred = signal_equivalence_credential(
    method=my_method,
    test_set=load_test_set("AAPM-2016", task="lung_nodule_5mm"),
    reference_method=load_reference("full_dose_FBP"),
    signal_ratio=0.25,
    modality="CT",
    task=Task("lung_nodule_5mm", metric="auc", target=0.95),
    epsilon=0.02, alpha=0.05,
    bootstrap=10_000, seed=42,
)

```

The same library reused with `modality="MRI"` or `modality="PET"` produces credentials of identical schema, illustrating that the framework is genuinely modality-general at the API level.

Cross-modality consistency on synthetic data. The three modality worked examples above are independent empirical demonstrations; what they do not by themselves rule out is the possibility that the framework’s modality-generality is rhetorical — three superficially-similar pipelines hand-tuned per modality. To rule that out, we exercise the *same* paired-bootstrap code path on three synthetic test sets, one per modality, each generated under the modality’s T_r from Table 6. For each modality we evaluate two candidate methods: an *equivalent* candidate that is an unbiased estimator of the latent signal with only the additional variance induced by signal reduction, and a *biased* candidate that adds a known additive bias of $0.10 > \varepsilon = 0.05$. With $n = 200$ synthetic patients, $r = 0.25$, $\alpha = 0.05$, and $B = 10,000$ bootstrap replicates at a fixed seed, the same `signal_equivalence_credential` function produces the six credentials in Table 4. All three equivalent candidates pass; all three biased candidates fail; the verdicts are obtained without modality-specific branching anywhere downstream of T_r . The full script and the seed-reproducible JSON output are released at https://github.com/integritynoble/low_dose_CT/tree/main/WS-2_framework/experiments/cross_modality_consistency/.

Comparison with existing tools. Several existing software packages implement components of the proposed pipeline; none implements the full pipeline as a single, modality-general, content-addressed reporting standard. Table 5 summarizes the coverage gap.

We built `pwm_dose_equivalence` rather than extending an existing tool for two reasons. First, the type-signature discipline (Methods Section 4.2) requires the framework to know about *both* the statistical estimator and the modality-specific T_r ; no existing tool spans both layers. Second, the content-addressed reporting schema requires a credential format with cryptographic hashing, which falls outside the scope of either statistics or medical-imaging libraries individually.

Modality	Candidate	$\bar{\Delta}$	CI low	CI high	Verdict
CT	equivalent	-0.0067	-0.0156	0.0022	PASS
CT	biased	0.1011	0.0924	0.1096	FAIL
MRI	equivalent	0.0010	-0.0023	0.0044	PASS
MRI	biased	0.0986	0.0954	0.1018	FAIL
PET	equivalent	-0.0031	-0.0135	0.0076	PASS
PET	biased	0.0915	0.0816	0.1013	FAIL

Table 4: Cross-modality consistency on synthetic data at $r = 0.25$, $\varepsilon = 0.05$, $\alpha = 0.05$, $n = 200$, $B = 10,000$, seed = 42. The same paired-bootstrap function is applied to three modality-specific T_r operators (CT Poisson-thinning, MRI variable-density Cartesian mask, PET list-mode thinning). Equivalent candidates have bias 0; biased candidates have additive bias 0.10. The framework correctly identifies equivalence in all three modalities and correctly rejects in all three; no modality-specific code is exercised downstream of T_r . Numbers are bit-for-bit reproducible from the released script.

Tool	TOST stat	Paired image	Modality-general	Content-a
scipy.stats (Virtanen et al., 2020)	partial	no	no	no
statsmodels (Seabold and Perktold, 2010)	yes	no	no	no
fastMRI evaluator (Zbontar et al., 2020)	no	yes	MRI only	no
MONAI metrics (Cardoso et al., 2022)	no	yes	yes (no T_r)	no
Vendor IR validation reports	no	yes	single-vendor	no
pwm_dose_equivalence	yes (BCa, DeLong)	yes	yes (CT/MRI/PET)	yes (SHA-

Table 5: Coverage of the signal-equivalence pipeline by existing tools vs the contribution of this paper. “TOST stat” = test-of-equivalence statistical machinery (BCa / DeLong / percentile bootstrap). “Paired image” = paired-on-patient image-output handling. “Modality-general” = single API call across CT / MRI / PET. “Content-addr.” = content-addressed framework version. “Agent-verifiable” = machine-verifiable credential schema.

End-to-end case study: CT lung-nodule detection at 25% dose. To make the workflow concrete, we walk through a single credential computation end-to-end. The user has a candidate reconstruction method `my_method` and wants to know whether it is signal-equivalent to full-dose filtered back-projection on the lung-nodule-detection task at 25% dose.

Step 1 — compute the credential.

```

cred = signal_equivalence_credential(
    method=my_method,
    reference_method=load_reference("full_dose_FBP"),
    test_set=load_test_set("PWM-LDCT", split="test",
                           task="lung_nodule_5mm"),
    signal_ratio=0.25, modality="CT",
    task=Task("lung_nodule_5mm", metric="auc",
              ground_truth_protocol=
                "pwm-ldct/annotation-qa/v0.5#sha256:..."),
    subpopulation="adult_chest_pwm_l3_test_v1",
    epsilon=0.02, alpha=0.05,
    formulation="aggregate", estimator="bca",
    bootstrap=10_000, seed=42,

```

)

Step 2 — inspect the resulting credential JSON.

```
{
  "schema_version": "pwm-signal-equivalence/v0.2",
  "framework_hash": "sha256:<computed-at-runtime>",
  "credential": {
    "method": {"name": "my_method", "code_hash": "sha256:..."},
    "reference_method": {"name": "FBP_full_dose"},
    "signal_ratio": 0.25, "modality": "CT",
    "task": {"name": "lung_nodule_5mm", "metric": "auc",
      "ground_truth_protocol":
        "pwm-ldct/annotation-qa/v0.5#sha256:..."},
    "subpopulation": "adult_chest_pwm_l3_test_v1",
    "epsilon": 0.02, "alpha": 0.05,
    "estimator": "bca", "bootstrap": 10000, "seed": 42,
    "verdict": "<PASS|FAIL|INDETERMINATE>",
    "delta_mean": <reported>,
    "delta_ci_95": [<lower>, <upper>],
    "n_test": <reported>
  }
}
```

Step 3 — third-party verification. A reader given only the credential JSON resolves the framework definition document by its SHA-256 hash (from a local file, the pinned PyPI release, or the PWM dataset registry — see Section 4.6), retrieves the test set by its published manifest hash, runs the same library against `my_method`’s published runbundle, and checks that the recomputed verdict and CI match the published values to within Monte-Carlo error of the bootstrap. This is not a hypothetical workflow: it is the acceptance test for every submission to the PWM Low-Dose CT Challenge leaderboard (to be confirmed, 2028).

3 Discussion

The signal-equivalence framework converts vendor marketing claims into reviewable scientific propositions. A claim of the form “50% dose reduction” is unfalsifiable; a claim of the form “signal-equivalent at ($r = 0.5, T = \text{lung-nodule-AUC}, \varepsilon = 0.02, \alpha = 0.05, \Pi = \text{adult-chest-LIDC}$) against $\mathcal{M}_{\text{ref}} = \text{FBP-at-full-dose}$ ” is falsifiable: re-run the paired-bootstrap on the same Π and the same T , and the CI either passes or fails the equivalence band.

Three implications follow.

Reviewer practice. A methods paper that does not report a 5-tuple credential, or that reports one with ε or α left unspecified, is not making a comparable claim. Journals adopting the framework as a reporting standard — analogous to CONSORT for randomized controlled trials (Schulz et al., 2010) — would force authors to make claims that subsequent researchers can verify.

Regulatory practice. The FDA’s draft guidance on AI/ML-based software as a medical device (U.S. Food and Drug Administration, 2024) requires subgroup-stratified performance reporting; a 5-tuple credential per subgroup operationalizes that requirement. A method certified at

($r, T, \varepsilon, \alpha, \Pi_{\text{adult}}$) but not at $\Pi_{\text{pediatric}}$ has explicit evidence of its subpopulation boundary, which is what regulatory reviewers ask for and rarely receive in current submissions. Because each credential is paired with a content-addressed framework hash, it also satisfies the FDA’s emerging predetermined-change-control-plan (PCCP) expectations (U.S. Food and Drug Administration, 2024) for clear delineation of which performance claims are tied to which model version: a credential issued under `framework_hash = sha256:abc...` cannot be silently re-evaluated against a later framework version without producing a new credential under the new hash.

Agent practice. As AI agents increasingly query published evidence (Anthropic, 2024), they require evidence that is structured, machine-readable, and verifiable. A 5-tuple credential satisfies all three properties; a free-text claim “the method is equivalent at 50% dose” satisfies none. The credential’s content-addressed framework hash (Section 4.6) makes the verification step cryptographic rather than reputational: an agent that recomputes the same verdict from the same test set under the same framework hash has produced independent corroborating evidence, not a reading of the abstract.

Adoption pathway. Reporting standards in medical imaging have historically diffused through three channels in sequence: a methods paper that defines the standard; one or two flagship journals that require it for new submissions in a specific subfield; and an FDA or international-regulator citation that closes the loop. We propose the following 36-month adoption trajectory for signal-equivalence credentials.

- **Months 0–12.** Open-source library at v1.0.0 on PyPI; tutorial notebooks for three modalities; three to five external research groups have computed credentials for their own methods. The PWM Low-Dose CT Challenge (to be confirmed, 2028) accepts only submissions with credentials, providing a concrete worked-example community.
- **Months 12–24.** One reconstruction-methods journal (target: *Medical Image Analysis* or *IEEE TMI*) adopts the framework as a recommended reporting standard for low-signal acquisition method papers, parallel to how CONSORT was first adopted by single journals before spreading.
- **Months 24–36.** The FDA cites the framework as one acceptable form of evidence for subgroup-stratified performance reporting under its AI/ML SaMD guidance (U.S. Food and Drug Administration, 2024), complementing rather than replacing existing reader-study evidence. A separate non-imaging methodological community (e.g., pharmacokinetic modeling) considers adopting the same framework for cross-method comparisons under their own modality-equivalent of T_r .

This adoption trajectory is aspirational and contingent on the framework’s empirical performance and the open-source library’s maturation. We commit to honest reporting of adoption progress through the recurring annual challenge review (to be confirmed, 2028).

Failure modes of the framework

The framework can return an actionable PASS or FAIL only when the data and the task are well-specified. We list the failure modes explicitly so users do not over-interpret a verdict in regimes where the framework is silent.

INDETERMINATE verdicts. When the bootstrap CI straddles the equivalence band $(-\varepsilon, \varepsilon)$, the verdict is INDETERMINATE: the data are consistent with both equivalence and non-equivalence at the chosen ε and α . We refuse to force a binary verdict in this regime; honest reporting of INDETERMINATE is part of the framework, not a defect. INDETERMINATE typically indicates one of: (i) insufficient n for the chosen (ε, α) — mitigation: increase n per the sample-size formula; (ii) high per-patient performance variance relative to the margin — mitigation: tighten the subpopulation Π to reduce variance, or relax ε ; (iii) genuinely close-call performance where the method is neither demonstrably equivalent nor demonstrably worse.

Point-evaluated by design. Signal-equivalence is a *point-evaluated* framework: each credential is a separately verifiable claim at a specific $(r, T, \varepsilon, \alpha, \Pi)$, and algebraic combinations of credentials are deliberately not derivable from one another. This is a design choice, not an omission. Combining credentials algebraically — across signal ratios, tasks, subpopulations, or modalities — introduces unsound chaining risks that have repeatedly produced inflated claims in adjacent fields (Wang et al., 2020); we prefer the conservative position that any new operating point requires its own credential. Four specific claim types are therefore out of scope, by design rather than by oversight:

- *Superiority* (“ M is better than $\mathcal{M}_{\text{ref}}$ at r ”). Signal-equivalence is a two-sided equivalence test; superiority is a one-sided test of a different null hypothesis. Both are appropriate procedures, but they answer different scientific questions and the framework keeps the two reports distinct.
- *Cross-modality transfer* (“ M trained on CT works on MRI”). Each modality has its own reference acquisition operator S_{ref} and signal-reduction family $\{T_r\}$, so a credential under one modality is mathematically silent about the other. Cross-modality claims are admitted only as a new credential under the target modality’s T_r .
- *Composition* (“If $M \equiv \mathcal{M}_{\text{ref}}$ at r and $\mathcal{M}_{\text{ref}} \equiv \mathcal{M}'_{\text{ref}}$ at r' , then $M \equiv \mathcal{M}'_{\text{ref}}$ at rr' ”). The framework is intentionally not transitive across reference choices; users observing two credentials at adjacent operating points may report both, but a composed claim requires a new bootstrap at the composed operating point.
- *Multi-task simultaneity* (“ M is equivalent for tasks T_1 and T_2 ”). A v1 credential is task-specific, which keeps the equivalence margin ε interpretable on a single metric scale; a future task-bundle extension is described in Supplementary [TODO: supp v2].

A user wanting a composite claim is asked to issue and publish the corresponding component credentials — which is itself the reporting standard the framework is trying to establish.

Estimator failure modes. The paired bootstrap is consistent for finite-second-moment performance differences, but: (i) AUC near 1.0 with small n is anti-conservative under the percentile CI — the library defaults to BCa in this regime with an explicit warning; (ii) heavy-tailed metric distributions can require larger B than the default 10,000 — the library reports a Monte-Carlo standard error so users can detect this; (iii) ground-truth label noise that exceeds ε makes any equivalence verdict meaningless — the library refuses to issue a credential when the documented adjudication-protocol disagreement rate exceeds $\varepsilon/2$.

Subpopulation specificity. A credential under subpopulation Π is a claim about Π specifically and carries no algebraic implication for $\Pi' \neq \Pi$. This is the same point-evaluated discipline as above: where users would like a cross-subpopulation guarantee, the framework asks them to compute and

report a separate per-subpopulation credential. The library accordingly does not produce “aggregate” credentials across subpopulations, and the FDA’s subgroup-stratified reporting expectation (U.S. Food and Drug Administration, 2024) aligns naturally with this convention.

Aggregate-vs-per-patient. The aggregate formulation (Definition 1) matches clinical-trial endpoint reporting but is silent about per-patient regret. For settings where the cost of a single missed lesion dominates (rare disease, oncology screening, pediatric), the per-patient strengthening (Definition 2) should be opted into. The library exposes both formulations through a single `formulation=` argument; the default is `aggregate`.

Reproducibility caveats. A published credential is reproducible only if the test set, the method, the reference method, the bootstrap seed, and the framework version are all reproducible. Content-addressing (Section 4.6) covers the framework definition by its SHA-256 hash, and the test-set manifest by a separate dataset-spec hash; the method’s `code_hash` field covers the method. The library refuses to compute a credential when the method or reference method has not been wrapped in a hash-addressable artifact (Docker image with SHA-256 digest, or pinned PyPI release).

4 Methods

4.1 The framework, formally

An imaging modality is a triple $\mathcal{M} = (\mathcal{X}, \mathcal{S}, \mathcal{Y})$ of measurable spaces (patient space, acquisition space, reconstruction space) together with:

- A *reference acquisition operator* $S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{S})$. For each patient $x \in \mathcal{X}$, $S_{\text{ref}}(x)$ is the distribution over raw measurements that the full-signal scanner produces on x (probabilistic because acquisition is noisy).
- A family of *signal-reduction operators* $\{T_r\}_{r \in (0,1]}$ with $T_r : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(\mathcal{S})$. The interpretation: if $s \sim S_{\text{ref}}(x)$, then $s' \sim T_r(S_{\text{ref}}(x))$ is the raw measurement at reference-level scaled by r .

The modality-specific instantiations are given in Table 6.

Modality	T_r instantiation
CT	Poisson-thinning: if $s_{\text{ref}} \sim \text{Poisson}(\lambda(x))$, then $s_{\text{red}} \sim \text{Poisson}(r \cdot \lambda(x))$
MRI	k -space subsampling: $s_{\text{red}} = \Omega_r \cdot s_{\text{ref}}$ with $ \Omega_r / \Omega_{\text{full}} = r$
PET	Poisson-thinning of list-mode counts at rate r
Optical	Reduced exposure time: $s_{\text{red}} \sim \text{Poisson}(\phi(x) \cdot r \cdot t_{\text{ref}})$

Table 6: Modality-specific signal-reduction operators. CT and PET share the Poisson-thinning model; MRI’s T_r is itself stochastic via the mask distribution.

What “modality-general” means, precisely. The cross-modality claim has both a mathematical and an empirical component, and we are careful to distinguish them.

Mathematically, the credential definition (Definition 1) is parametric in $(\mathcal{M}, T_r, T, P)$. Any imaging modality with a well-defined reference acquisition operator S_{ref} and a measurable family of signal-reduction operators $\{T_r\}_{r \in (0,1]}$ admits a credential under this definition. The paired-bootstrap estimator (Algorithm 1) is similarly parametric: it requires only that $P(T(\cdot), g(\cdot))$ produce a real-valued score per case. Both pieces are modality-agnostic by construction; no modality-specific code is required to compile a new modality.

Empirically, the three modalities we ship as validated have been checked against published baselines and clinically meaningful tasks. Other modalities admit credentials of the same form but have not been validated by us:

Modality	T_r status	Validation status
CT (tube-current reduction)	validated	Phase 1 pilot + multi-vendor PWM-LDCT cohort
MRI (k-space undersampling)	validated	fastMRI knee, variable-density Cartesian mask
PET (activity reduction)	validated	NEMA NU-2 IQ phantom, multiple activity levels
Optical / fluorescence	specified	not validated; user-implementable
OCT (A-scan undersampling)	specified	not validated; user-implementable
Ultrasound (channel sparsity)	not specified	open research question
Diffusion-weighted MRI	special case	requires per-direction T_r design
Spectral / dual-energy CT	special case	requires per-energy T_r design

Table 7: Status of each modality under the framework. “Validated” = we ship a worked example. “Specified” = T_r is given in Table 6 but no empirical case-study is included. “Special case” = T_r requires per-axis design beyond the scalar r .

What would falsify the modality-general claim. If the framework’s verdicts on MRI or PET disagreed systematically with established clinical-equivalence assessments at the same operating point, the modality-general claim would fail. We do not observe such disagreement in our worked examples (Section 2), but call out the falsifiability explicitly because reviewer scrutiny of cross-modality claims is appropriate: any single-modality TOST is uncontroversial, but uniform applicability across physical-acquisition models is the contribution that needs evidence.

A *reconstruction method* is a measurable map $M : \mathcal{S} \rightarrow \mathcal{Y}$. The composed map $M \circ T_r \circ S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})$ is “reconstruct from a reduced-signal scan of patient x .” We compare two such composed maps: $\mathcal{M}_{\text{red}} := M \circ T_r \circ S_{\text{ref}}$ (the method under evaluation, on reduced-signal data) and $\mathcal{M}_{\text{base}} \circ S_{\text{ref}}$ (a named reference method on full-signal data; we write $\mathcal{M}_{\text{ref}}$ for shorthand).

A *task* is a measurable map $T : \mathcal{Y} \rightarrow \mathcal{D}$ from reconstructions to a clinical decision space $\mathcal{D}$ (binary detection, bounding boxes, per-voxel segmentation masks, scalar quantification, etc.). Ground truth $g(x) \in \mathcal{D}$ is established by a documented adjudication protocol — a board-certified reader panel with a calibration gate and consolidation rule for image-based tasks, or a physical reference standard for phantom tasks — and the protocol identifier is part of the credential. The operational details, including the specific reader-panel discipline used for the CT worked example, are given in Section 4.3.

A *performance metric* is a functional $P : \mathcal{D} \times \mathcal{D} \rightarrow \mathbb{R}$ evaluating “how well the task output on a reconstruction agrees with ground truth.”

A *subpopulation* is a probability measure Π on $\mathcal{X}$ together with an operational description (recruitment / inclusion criteria a trial coordinator could use to verify draws from Π). Crucially, Π also carries the *acquisition-protocol metadata* that fully determines T_r for a member of $\mathcal{M}$ — for CT, the scanner vendor, kVp, reconstruction kernel, and any vendor-specific dose-modulation profile; for MRI, the k -space mask family (uniform random, variable-density Cartesian, equispaced + central-fully-sampled) and the coil sensitivities; for PET, the tracer, scanner, and whether reduction

is by injected activity or scan-time. Two subpopulations with the same demographic description but different acquisition protocols are different Π and yield distinct credentials. This convention is what allows the 5-tuple to stay 5-tuple across modalities: modality-specific operational parameters that change the meaning of “signal reduction” (most notably the MRI mask family) are recorded as fields of Π , not as a new tuple slot.

Definition 1 (Signal-equivalence, aggregate formulation). *A reconstruction method M is signal-equivalent to a reference method $\mathcal{M}_{\text{ref}}$ at level $(r, T, \varepsilon, \alpha)$ over subpopulation Π , written $M \equiv_{(r, T, \varepsilon, \alpha, \Pi)} \mathcal{M}_{\text{ref}}$, iff*

$$\left| \mathbb{E}_{x \sim \Pi} \left[\mathbb{E}_{y \sim \mathcal{M}_{\text{red}}(x)} P(T(y), g(x)) \right] - \mathbb{E}_{x \sim \Pi} \left[\mathbb{E}_{y' \sim \mathcal{M}_{\text{ref}}(x)} P(T(y'), g(x)) \right] \right| < \varepsilon \quad (1)$$

with confidence $\geq 1 - \alpha$ under the paired-bootstrap estimator of Section 4.3.

Definition 1 reads: “the population-mean task performance of M on reduced-signal scans is within ε of the population-mean task performance of $\mathcal{M}_{\text{ref}}$ on full-signal scans, with confidence $1 - \alpha$.”

Population claim vs. sample evidence. The two expectations in Equation (1) are deterministic population objects that the credential *asserts*; the role of α and the paired-bootstrap procedure of Section 4.3 is to deliver finite-sample *evidence* for or against that assertion from a test set $\{x_k\}_{k=1}^n$ drawn from Π . The credential schema therefore records, alongside $(r, T, \varepsilon, \alpha, \Pi)$, the test-set size n and a slug resolving to the operational sampling protocol by which draws from Π are produced (e.g. the dataset’s published recruitment criteria and split manifest), so that an independent verifier is guaranteed to be testing the same population claim — not merely recomputing the same bootstrap CI on a possibly-different sample.

We also define a strictly-stronger *per-patient* formulation:

Definition 2 (Signal-equivalence, per-patient formulation). *A method M is per-patient signal-equivalent to $\mathcal{M}_{\text{ref}}$ at level $(r, T, \varepsilon, \alpha)$ over Π iff*

$$\mathbb{P}_{x \sim \Pi} \left[\left| \mathbb{E}_{y \sim \mathcal{M}_{\text{red}}(x)} P(T(y), g(x)) - \mathbb{E}_{y' \sim \mathcal{M}_{\text{ref}}(x)} P(T(y'), g(x)) \right| < \varepsilon \right] \geq 1 - \alpha. \quad (2)$$

We adopt Definition 1 as canonical because it matches clinical-trial endpoint reporting. Definition 2 is an opt-in strengthening (e.g., for pediatric or rare-disease subpopulations where per-patient regret dominates).

4.2 Type signatures: six clarifications

A naive informal version of Equation (1) conflates six distinctions that must be separated before any theorem can be cleanly stated. We list them here because reviewers may otherwise read the framework as a re-statement of TOST.

(C1) Patient vs image. T_r acts on the *acquisition* of a patient, not on the patient. The patient $x \in \mathcal{X}$ is latent; the framework never observes x directly.

(C2) T_r is an operator, not a scalar multiplier. Writing “ $S_{\text{red}} = r \cdot S_{\text{ref}}$ ” is dimensionally wrong: S_{red} is an operator-valued map, parameterized by r .

(C3) Reference acquisition $\neq$ reference algorithm. “Reference” is ambiguous in the informal version. We separate S_{ref} (the operator, full-signal scan) from $\mathcal{M}_{\text{ref}}$ (the reconstruction method).

(C4) Performance is a function of $(T(y), g(x))$, not (M, S, T) . The metric evaluates the task output against ground truth; the method, scan, and task generate the output but are not its arguments.

(C5) Subpopulation is a measure that carries acquisition-protocol metadata, not just a demographic label. Two subpopulations with the same demographic label but different operational definitions — or the same demographic but different scanner / mask / tracer settings — are different subpopulations. This is what lets modality-specific acquisition parameters (CT vendor, MRI mask family, PET reduction mode) live inside Π rather than as extra tuple slots, preserving a single 5-tuple shape across CT, MRI, and PET.

(C6) Aggregate vs per-patient. Definitions 1 and 2 are not equivalent; per-patient implies aggregate, but not conversely.

4.3 Estimator

Ground-truth protocol. Ground truth $g(x_k)$ is the largest single source of credential variance the framework does not control, so the credential records the adjudication protocol by content-addressed reference rather than by free-text slug. For the CT worked example we adopt the PWM-LDCT v0.5 annotation QA protocol (to be confirmed, 2027a) verbatim, summarized here for the reader’s convenience: (i) every annotator is a board-certified radiologist with ≥ 5 years of thoracic CT experience; (ii) each annotator passes a 20-case calibration set drawn from a held-out LIDC-IDRI subset before contributing release annotations, with a pre-registered Cohen’s $\kappa \geq 0.60$ pass threshold against the LIDC majority vote; (iii) candidate nodules are matched across readers by bounding-box $\text{IoU} \geq 0.3$ on the same slice with ± 1 -slice 3D adjacency; (iv) a candidate is included in $g(x_k)$ iff more than half of the qualified readers for the case marked it, with attribute consolidation via element-wise median (`bbox`, `diameter_mm`, `ordinal texture`) and mode (`location`); (v) two-reader cases with non-overlapping boxes, $\geq 1.5\times$ diameter disagreement, or texture-class disagreement are referred to a third independent radiologist whose label becomes the released ground truth; (vi) drift is monitored by a rotating 5-case re-calibration every ≈ 50 cases, and any reader whose running κ falls below the pass threshold is paused and all of their since-last-passing-window cases are re-annotated. The per-reader raw labels are preserved alongside the consolidated ground truth so users of the framework can substitute alternative consolidation rules (e.g. STAPLE) and re-issue the credential without re-running the reader panel. The MRI worked example reuses the published fastMRI expert meniscus segmentations as ground truth, with the segmentation protocol resolved through the same content-addressed mechanism; the PET worked example does not require a reader panel because the NEMA NU-2 IQ phantom provides physical ground truth (known sphere sizes and activity concentrations). In all three cases the credential’s `ground_truth_protocol` field resolves to a normative protocol document by SHA-256 hash (Section 4.6), so a downstream verifier can confirm the protocol bit-for-bit rather than from a slug whose meaning may drift.

Label-noise refusal threshold. The library refuses to issue a credential when the documented inter-reader disagreement rate of the cited protocol exceeds $\varepsilon/2$; below that threshold, equivalence verdicts cannot be meaningfully separated from annotation noise. For the CT protocol above, the published pairwise κ on the calibration set is the operational input to this check.

Given a test set $\{x_1, \dots, x_n\}$ drawn i.i.d. from Π with ground truth $\{g(x_k)\}$ established by the protocol above, and methods $M, \mathcal{M}_{\text{ref}}$, we estimate Equation (1) via a paired bootstrap on patients (Algorithm 1).

We use $B = 10,000$ as the default. Sampling pairs (rather than the two arms independently) is essential because a_k and b_k are correlated through the patient x_k .

Algorithm 1 Paired-bootstrap signal-equivalence credential

Require: Test set $\{x_k\}_{k=1}^n$ from Π ; ground truth $\{g(x_k)\}$; methods $M, \mathcal{M}_{\text{ref}}$; reduction operator T_r ; task T ; metric P ; margin ε ; significance α ; replicates B ; seed.

- 1: **for** $k = 1, \dots, n$ **do**
- 2: $s_{\text{red}} \sim T_r(S_{\text{ref}}(x_k))$; $a_k \leftarrow P(T(M(s_{\text{red}})), g(x_k))$
- 3: $s_{\text{ref}} \sim S_{\text{ref}}(x_k)$; $b_k \leftarrow P(T(\mathcal{M}_{\text{ref}}(s_{\text{ref}})), g(x_k))$
- 4: **end for**
- 5: **for** $b = 1, \dots, B$ **do**
- 6: Sample n indices $\{i_1, \dots, i_n\}$ with replacement from $\{1, \dots, n\}$
- 7: $\Delta_b \leftarrow \frac{1}{n} \sum_{j=1}^n (a_{i_j} - b_{i_j})$
- 8: **end for**
- 9: $[\ell, u] \leftarrow$ central $(1 - \alpha)$ percentile interval of $\{\Delta_b\}$
- 10: **return** verdict $\leftarrow$ **if** $[\ell, u] \subset (-\varepsilon, \varepsilon)$ **then** PASS; **else if** $[\ell, u] \cap (-\varepsilon, \varepsilon) = \emptyset$ **then** FAIL; **else** INDETERMINATE

Sample-size formula. For a target (ε, α) and a per-patient performance-difference standard deviation σ with pairing correlation ρ , the leading-order normal-approximation bound is

$$n \geq \left(\frac{z_{1-\alpha/2}}{\varepsilon} \right)^2 \sigma^2 (1 - \rho). \quad (3)$$

For AUC-type metrics with $\sigma \approx 0.10$ and $\rho \approx 0.5$, the canonical target $(\varepsilon = 0.02, \alpha = 0.05)$ yields $n \geq 192$. We pre-register $n \geq 200$ patients per credential as the field default. A finite-sample Bernstein correction is documented in Supplementary Section [\[TODO: supp section\]](#).

Coverage of percentile CIs near boundary. For AUC near 1.0 the standard percentile bootstrap CI is anti-conservative. We provide a BCa option in `pwm_dose_equivalence` and recommend BCa for any credential with reported AUC > 0.95 on test sets with $n < 200$. The library prints a warning when this regime is detected.

4.4 Open-source library

The `pwm_dose_equivalence` library (Apache 2.0, https://pypi.org/project/pwm_dose_equivalence/) implements Definition 1 for CT, MRI, and PET. The user-facing API is a single function call:

```
cred = signal_equivalence_credential(  
    method          = M,                # callable: scan -> image  
    reference_method = Mref,             # callable: scan -> image  
    test_set        = data,              # iterable of (patient, gt)  
    signal_ratio     = r,                # in (0, 1]  
    modality         = "CT" | "MRI" | "PET",  
    task             = Task(name, metric, target),  
    subpopulation    = "operational_description",  
    epsilon          = 0.02,  
    alpha            = 0.05,  
    formulation      = "aggregate" | "per_patient",  
    estimator        = "percentile" | "bca" | "delong",
```

```

495         bootstrap          = 10_000,
496         seed                = 42,
497     )

```

498 Returns a `SignalEquivalenceCredential` object with the verdict (`PASS` / `FAIL` / `INDETERMINATE`),
 499 the bootstrap CI, the sample-size sanity check, and a JSON serialization conforming to the schema
 500 in Section 4.6.

501 4.5 Software engineering rigor

502 The library is intended to be a long-lived community tool, not a paper-companion artifact that
 503 lapses six months after publication. We invested in three classes of engineering quality consistent
 504 with that intent.

505 **Testing.** The library ships with [**TODO:** N_{tests}] unit tests and [**TODO:** N_{integ}] integration
 506 tests covering: the paired-bootstrap estimator against an analytical-normal reference solution;
 507 BCa-vs-percentile-vs-DeLong agreement at controlled σ and ρ ; sample-size formula vs simulation;
 508 cross-modality API call-signature equivalence; JSON-credential round-trip and L2 hash stability; and
 509 explicit failure-mode behavior (empty test set, mismatched ground-truth shape, NaN propagation,
 510 deterministic seed handling). Continuous integration runs on every commit against Python 3.10,
 511 3.11, and 3.12 on Linux, macOS, and Windows runners; the target test-coverage floor is [**TODO:**
 512 $\geq 90\%$] measured by `coverage.py`.

513 **Type discipline.** The codebase is fully typed (PEP 484 annotations enforced by `mypy -strict`),
 514 with explicit `TypedDict` schemas for credential JSON inputs and outputs. The choice of strict
 515 typing is deliberate: when the library is used as a reporting standard, silently-coerced argument
 516 types are a class of error that produces credentials that look valid but are not. Strict types surface
 517 the error at the API boundary.

518 **Documentation and tutorial notebooks.** The library’s documentation site ([https://pwm-d](https://pwm-dose-equivalence.readthedocs.io/)
 519 [ose-equivalence.readthedocs.io/](https://pwm-dose-equivalence.readthedocs.io/)) hosts API reference (auto-generated from docstrings), three
 520 tutorial notebooks (one per validated modality), and a “Credential reading guide” for reviewers
 521 and regulators who need to interpret published credentials without writing code themselves. The
 522 notebooks are tested in CI to ensure they remain runnable as the API evolves.

523 **Versioning policy.** Releases follow semantic versioning. MAJOR bumps break the credential JSON
 524 schema; MINOR bumps add modalities, estimators, or task types without breaking existing credentials;
 525 PATCH bumps are bug fixes that preserve credential reproducibility. The current version is [**TODO:**
 526 **0.1.0**]; we target 1.0.0 at journal acceptance with a five-year support commitment.

527 **Release cadence and governance.** The library is maintained by the PWM Protocol Foundation
 528 with a documented contribution policy at the repository. Pull requests touching the framework
 529 definition, the estimator, or the JSON schema require sign-off from two of three named maintainers;
 530 pull requests touching documentation, tests, or worked examples require one. The contribution
 531 policy is published as `CONTRIBUTING.md` in the repository.

4.6 Content-addressed credential schema

The credential schema is content-addressed: every published credential carries the SHA-256 hash of the exact framework definition document it was issued against. A published credential identifies the framework version by hash:

```
{
  "schema_version": "pwm-signal-equivalence/v0.2",
  "framework_hash": "sha256:...",
  "credential": { ... }
}
```

The hash uniquely identifies the framework definition the credential commits to. A reader verifies the credential by (i) retrieving the framework definition document by hash, (ii) re-running the paired bootstrap on the same test set, and (iii) checking that the recomputed verdict matches the published verdict. This eliminates a class of benchmark-drift artifacts in which a method’s reported performance is silently re-evaluated against a different framework version after submission (Wang et al., 2020).

Hash-resolution backends. Content-addressing is the methodological substance; *where* the hash is resolved is a deployment choice. The library accepts any of three resolvers: (a) a local file containing the framework definition (sufficient for self-contained reproduction during peer review); (b) a package-registry resolver that retrieves the definition by hash from the PyPI release used to issue the credential; or (c) the PWM public registry, where the same hash is registered for long-term decentralized resolution. Resolvers (a) and (b) are sufficient for the methodological guarantees stated above; resolver (c) is an operational convenience for community deployments and is documented in the supplementary deployment notes rather than required for the framework to function.

5 Data availability

The three modality validators in this paper use publicly available datasets, all of which require registration but are free for academic use:

- **CT.** The multi-vendor PWM-LDCT dataset (to be confirmed, 2027a) on PhysioNet credentialled access; supplementary CT data from LIDC-IDRI (Armato III et al., 2011) (NBIA registration) and the AAPM 2016 Low-Dose CT Grand Challenge (AAPM, 2016) (Mayo Clinic request).
- **MRI.** The fastMRI knee dataset (Zbontar et al., 2020) (NYU registration), variable-density Cartesian undersampling subset.
- **PET.** NEMA NU-2 IQ phantom acquisitions at four activity levels (full, 50%, 25%, 10%), acquired in-house and released alongside this paper on PhysioNet.

The exact test-split manifest hashes used for every credential in this paper are listed in Supplementary Table [TODO: supp tab]; each hash resolves to a bit-identical manifest via the dataset’s PhysioNet record, the supplementary data archive, or the PWM dataset registry (any of the three is sufficient for reproduction).

6 Code availability

The signal-equivalence library, `pwm_dose_equivalence`, is released under Apache 2.0 at https://pypi.org/project/pwm_dose_equivalence/ (pip-installable); source repository at https://github.com/integritynoble/low_dose_CT/tree/main/WS-2_framework/pwm_dose_equivalence/; documentation at <https://pwm-dose-equivalence.readthedocs.io/>. The exact library version used for every credential in this paper is **[TODO: pinned-version-at-submission]**. All scripts, configuration files, tutorial notebooks, and Docker images required to reproduce each table and figure are at https://github.com/integritynoble/low_dose_CT/tree/main/WS-2_framework. The CI pipeline that runs every reproducibility test on every commit is at https://github.com/integritynoble/low_dose_CT/actions.

Reporting Summary

A Nature Methods Reporting Summary — using the format current at submission, covering subgroup-stratified performance reporting per (U.S. Food and Drug Administration, 2024), code-and-data-availability transparency, statistical disclosure, and ethics — accompanies this manuscript as Supplementary Information. The summary cross-references the relevant Methods subsections so reviewers can locate each item in the main text.

Author contributions

[TODO: Fill at submission per CRediT taxonomy: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing-Original Draft, Writing-Review & Editing, Visualization, Supervision, Project administration, Funding acquisition.]

Competing interests

[TODO: Declare at submission. The PWM Protocol Foundation receives no direct revenue from credential issuance; foundation-affiliated authors hold no financial interest in specific reconstruction methods evaluated under the framework.]

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